

1. Problem Statement

Airbnb operates as a global online marketplace for short-term accommodation across multiple cities, offering various property types hosted by independent providers. Over time, the platform has experienced rapid growth, regional demand variations, pricing differences, and changes in customer behavior.

However, stakeholders lack a centralized analytical view to understand:

- Which cities contribute most to business growth
- How pricing and property types influence adoption
- Customer satisfaction and review behavior
- Seasonal demand patterns
- Trust and verification levels of hosts

To address these challenges, this project leverages Power BI-based data analytics to uncover actionable insights from listing, pricing, review, and host data, enabling Airbnb to make data-driven strategic decisions for growth, quality improvement, and customer trust.

2. Goal of the Dashboard

The primary goal of this dashboard is to:

- Provide a global overview of Airbnb performance
- Identify high-performing cities and markets
- Analyze pricing strategies and property preferences
- Evaluate customer satisfaction and ratings drivers
- Understand customer behavior through review patterns
- Detect seasonality trends in demand
- Assess host trust and platform safety
- Support business expansion, marketing, and operational decisions

3. Key Visuals in Dashboard

The dashboard includes the following key visuals:

- **KPI Cards**
 - Total Listings
 - Total Cities
 - Total Hosts
 - Property Types
 - Total Reviews
- **New Listings Trend (Time Series)**
 - Shows lifecycle: Introduction → Growth → Maturity → Decline → Reinvention → COVID impact
- **Market Share by City (Bar + Cumulative Line Chart)**
 - City-wise contribution to total listings
- **Property Type & Pricing (Bar Chart)**
 - Price comparison: Hotel vs Airbnb types
- **Ratings Heatmap (Matrix Visual)**
 - Metrics: Accuracy, Cleanliness, Communication, Location, Value
- **Review Frequency Distribution (Histogram + Line)**
 - Customer engagement behavior
- **Seasonality Analysis (Stacked Area Chart)**
 - Monthly review trends by city
- **Trust & Verification (Donut Chart)**
 - Verified vs Non-verified hosts

4. Insights (Based on Client Questions + Dashboard Data)

1. Big Picture / Overview

- Airbnb operates across 10 cities with 2,79,712 listings and 1,82,024 hosts
- The platform experienced peak growth in 2015, followed by slowdown due to regulations and COVID-19 impact
- Growth lifecycle shows clear phases: rapid growth → maturity → decline → recovery

👉 Insight: Airbnb is a mature platform with external dependency factors (regulations, pandemics)

2. Market Share by City

- Paris, New York, and Sydney contribute nearly 50% of listings and 59% of reviews
- Paris has the highest listings and reviews globally

👉 Insight:

- Market is highly concentrated, not evenly distributed
- Few cities dominate supply → risk of over-dependence

3. Property Type & Pricing

- Average prices:
 - Hotel Room: \$800
 - Entire Place: \$673
 - Shared Room: \$580
 - Private Room: \$462
- Hotels are significantly more expensive → drives Airbnb adoption

👉 Insight:

- Airbnb succeeds due to price advantage over hotels
- “Entire Place” is a strong premium segment

4. Ratings & Customer Satisfaction

- Best-rated cities: Mexico City, Rio de Janeiro
- Worst-rated cities: Hong Kong, Istanbul
- Lowest scoring factors:
 - Cleanliness
 - Value for money

👉 Insight:

- High supply ≠ high quality
- Operational quality issues directly impact ratings

5. Review Frequency (Customer Behavior)

- Most users write only 1 review
- 98.8% of users write ≤ 3 reviews

👉 Insight:

- Customer engagement is low and skewed
- Reviews are not frequent → limited feedback loop

6. Seasonality Analysis

- Paris & Rome peak: April – August (summer travel)
- New York peaks: Nov – Dec (holiday season)

👉 Insight:

- Demand is highly seasonal and location-dependent
- Travel patterns vary by geography

7. Trust & Host Verification

- 66.9% hosts fully verified
- Nearly all hosts provide at least one trust signal

👉 Insight:

- Platform maintains strong trust foundation
- Very low anonymous/unverified presence

5. Recommendations

1. Business Growth

- Expand in **under-penetrated cities** (reduce dependency on top 3 cities)
- Invest in **emerging markets (Asia, Latin America)**

2. Pricing Strategy

- Promote **"Entire Place" segment** for higher revenue
- Maintain **price advantage vs hotels**
- Introduce **dynamic pricing models** for seasonal demand

3. Customer Experience

- Improve:
 - Cleanliness standards
 - Value for money perception
- Introduce:
 - Host quality training
 - Standardization guidelines

4. Engagement Improvement

- Encourage more reviews via:
 - Incentives (discounts, coupons)
 - Simplified review process

5. Seasonality Optimization

- Launch **seasonal marketing campaigns**

- Help hosts with **demand forecasting tools**

6. Trust & Safety

- Increase **host verification to >80%**
- Highlight **trusted hosts in search ranking**
- Strengthen fraud detection

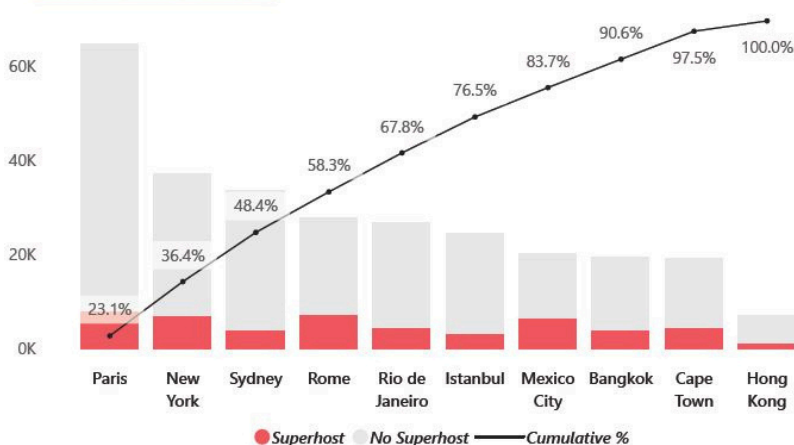
7. Risk Mitigation

- Reduce reliance on top cities
- Prepare contingency strategies for:
 - Regulatory changes
 - Global disruptions (like COVID)

Global Airbnb Performance Dashboard



Market Share by City



Paris, NYC, and Sydney account for almost half of the total listings and **59%** of total reviews.

Paris is the city with the **most listings and reviews**. A possible driver is the **prices** for hotel rooms **being twice** as much as Airbnb!

Hotel room	\$800
Entire place	\$673
Shared room	\$580
Private room	\$462

Avg Price

Select the detail level



Overall Rating



Detailed Rating

Ratings

city	Accuracy	Cleanliness	Communication	Location	Value
Bangkok	9.5	9.4	9.6	9.2	9.3
Istanbul	9.3	9.1	9.5	9.4	9.2
Hong Kong	9.2	9.0	9.4	9.6	9.0
Rome	9.6	9.5	9.7	9.6	9.4
New York	9.6	9.3	9.7	9.6	9.4
Paris	9.6	9.2	9.7	9.7	9.3
Sydney	9.5	9.2	9.7	9.7	9.3
Cape Town	9.6	9.5	9.7	9.7	9.5
Rio de Janeiro	9.6	9.4	9.8	9.8	9.3
Mexico City	9.7	9.6	9.8	9.8	9.6

Mexico City and *Rio* are the overall **best-rated cities**, *HK* and *Istanbul* the **worst ones**. **Cleanliness** and **value for money ratio** are the two metrics generally scoring the lowest.